

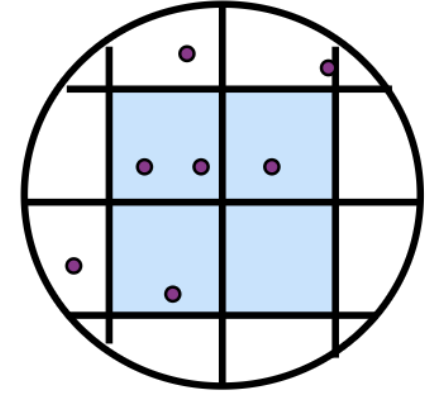
Semiconductors: New Technologies and the Value Chain

Module 6: Future Directions
Live Session

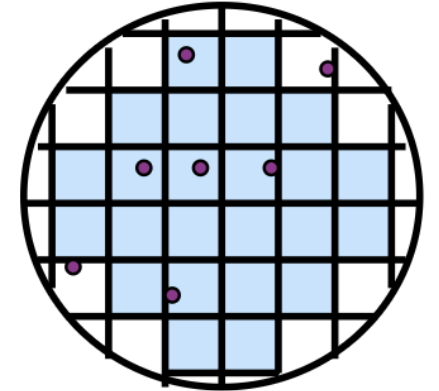
Questions from Previous Sessions

1. How does D0 yield translate to upstream and downstream in the value chain?
 - $\text{yield} \sim (\text{die area})^{-3} \rightarrow \text{cost} \sim (\text{die area})^4$
 - Engineering change order (ECO) that links upstream and downstream
 - upstream & midstream: design for manufacturability
2. What are buffers and how are they used in IC design?
 - Inverters are the most effective circuit in driving large capacitive loads.
 - Rule of thumb: exponential horn – chains of increasing sized inverters (1x – 2x – 4x)
3. How is packaging design impacted from the original design process due to miniaturisation of the chips?
 - Package dictates the floorplanning of I/O circuitries which scale much slower than the transistors.
 - Parallel vs series transfer, time-multiplexed pins

Yield = $1/4 = 25\%$



Yield = $19/24 = 79\%$



Learning Outcomes

By the end of this module you will be able to:

- Review the current status of semiconductor technologies in key application areas.
- Identify the main challenges in the application areas and evaluate feasible solutions.
- Evaluate the challenges in the semiconductor value chain holistically.

A warm-up question

What are the policies on semiconductor around the world?

- Economic dependency
- Dominance
- Resilience

Chip Act 2.0

- First Chips Act in 2023 unlocks €52 billion, creating 46,000 new jobs. ICs are specialised for autonomous vehicles, 6G telecoms infrastructure, industrial robots and drones [1].
- Industrial Alliance for Semiconductors reduce the region's dependence (currently 80%) on imported microchips and shock-prone supply chains.
- Geopolitics: steer away from US and work closer with Japan, South Korea, and Taiwan.
- Address shortage of talents: requires > 100,000 people to staff new factories. Workers to be trained in universities, pre-skilled and redeployed.

Breakout Group

- Take turns to discuss your thoughts:
"What should be the focus of semiconductor development in your region?"
- Time: 10 minutes

Final Assessment

- Free to choose your role
- Report / proposal / discussion paper for your stakeholders
- Correlate materials from modules 1-6 and demonstrate your knowledge across the semiconductor value chain
- Is any evidence from the practical tasks (Openlane) supportive to your discussion? If so, include them.

Sections	Length (words)
Context statement	100
Current Challenges	500
Technological solutions	500
Action plans	400

- Extend your learning into your work and inspire new ideas/opportunities.
- Less is more!

Questions and Answers

- Do you have any questions from today's discussion or the Canvas materials?

References

[1] <https://sciencebusiness.net/news/eus-semiconductor-strategy-evolves-new-era>

[2] <https://www.eetimes.com/beyond-the-fab-building-europe-next-generation-of-semiconductor-champions/>